

Astronomy C - Astronomy - Monta Vista Science Olympiad (Div. C) - 11-14-2025
Student Answer Sheet (/monta-vista/SM/AnswerSheet?tid=000GJX)

Instructions (shown before students start the test)

Introduction (shown after students start the test)

General Knowledge

1. (3.00 pts) Which process dominates the energy production in the core of a massive star nearing the end of its main-sequence life?

- ☐ A) Proton-proton chain
- ☐ B) CNO cycle
- ☐ C) Triple-alpha process
- ☐ D) Neutron capture

2. (3.00 pts) What is the primary driver of pulsations in a Cepheid variable star?

- ☐ A) Rotational instability
- ☐ B) Kappa mechanism
- ☐ C) Magnetic field oscillations
- ☐ D) Gravitational wave emissions

3. (3.00 pts) In the H-R diagram, which evolutionary track is characterized by a rapid transition from the main sequence to the red giant branch due to helium flash?

- ☐ A) Low-mass stars
- ☐ B) Intermediate-mass stars
- ☐ C) High-mass stars
- ☐ D) White Dwarfs

4. (3.00 pts) Which spectral line transition is most critical for studying the ionization state in H II regions?

- ☐ A) Lyman-alpha
- ☐ B) Balmer-alpha
- ☐ C) Paschen-alpha
- ☐ D) Brackett-alpha

5. (3.00 pts) What triggers the collapse of a white dwarf in a Type Ia supernova?

- ☐ A) Exceeding the Tolman-Oppenheimer-Volkoff limit
- ☐ B) Loss of magnetic field stability
- ☐ C) Core helium depletion
- ☐ D) Reaching the Chandrasekhar limit

6. (3.00 pts) Which of these best describes the chemical composition shift in a star's envelope during the asymptotic giant branch phase?

- ☐ A) Increase in hydrogen abundance
- ☐ B) Dominance of oxygen
- ☐ C) Enrichment with s-process elements
- ☐ D) Depletion of helium

7. (3.00 pts) Which of these is a feature of blackbody radiation?

- ☐ A) Discrete emission lines
- ☐ B) Continuous spectrum
- ☐ C) Only ultraviolet emission
- ☐ D) No thermal energy

8. (3.00 pts) Which observational feature distinguishes Mira variables from other red giants?

- ☐ A) Longer pulsation periods exceeding 100 days
- ☐ B) Higher surface temperature
- ☐ C) Lack of infrared excess
- ☐ D) Stable luminosity

9. (3.00 pts) What is the likely progenitor of a millisecond pulsar in a binary system?

- ☐ A) A neutron star spun up by accretion
- ☐ B) A white dwarf with rapid rotation
- ☐ C) A protostar with strong magnetic fields
- ☐ D) A black hole merger remnant

10. (3.00 pts) Which factor most significantly affects the initial mass function of stars forming in a molecular cloud?

(Mark **ALL** correct answers)

- ☐ A) Cloud temperature
- ☐ B) Magnetic field strength

- ☐ C) Metallicity gradient
- ☐ D) Cosmic ray flux

11. (3.00 pts) The _____ is a stellar pulsation driver caused by opacity changes due to ionization in the star's atmosphere.

12. (3.00 pts) H II regions are areas of _____ hydrogen surrounding young, massive stars, emitting _____ lines.

13. (3.00 pts) The initial evolutionary path of a low-mass protostar on the H-R diagram, moving vertically at constant temperature.

14. (3.00 pts) The maximum mass a neutron star can have before collapsing into a black hole. (The term for it, not a numerical answer)

15. (3.00 pts) Name a key isotope produced in Type Ia supernovae before any decay has occurred.

16. (3.00 pts) Late-stage evolved stars with helium burning shells and hydrogen shells, rich in s-process elements.

17. (3.00 pts) Electromagnetic radiation emitted when charged particles are decelerated by other particles' electric fields.

18. (3.00 pts) _____ luminosity is the maximum luminosity a star can achieve when radiation pressure balances gravity.

19. (3.00 pts) At a frequency of _____ MHz in the water hole, the emission for neutral H can be found.

20. (3.00 pts)

At masses greater than 1.5 solar masses, stars have two different types of heat transfer occurring within them: _____ in the core and _____ in the surrounding envelope

DSOS

Use the following images and answer their respective questions directly under them.



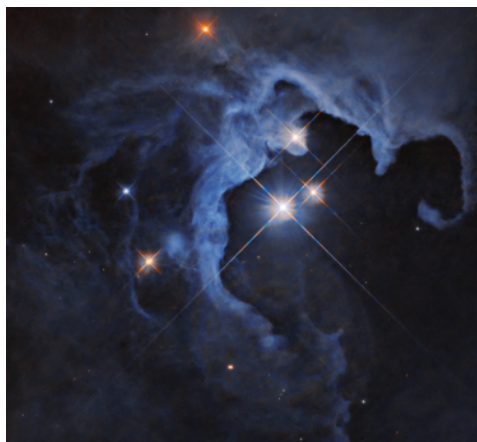
21. (3.00 pts) Identify the DSO in this image

22. (3.00 pts) Which wavelength was this image taken in?

23. (3.00 pts) What type of object is this DSO, and what does this term mean?

24. (3.00 pts) What type of star is at this object's center, and how is it involved in the object's emission mechanism?

25. (3.00 pts) This object is notable for its cometary knots. What are they and how were they likely formed?



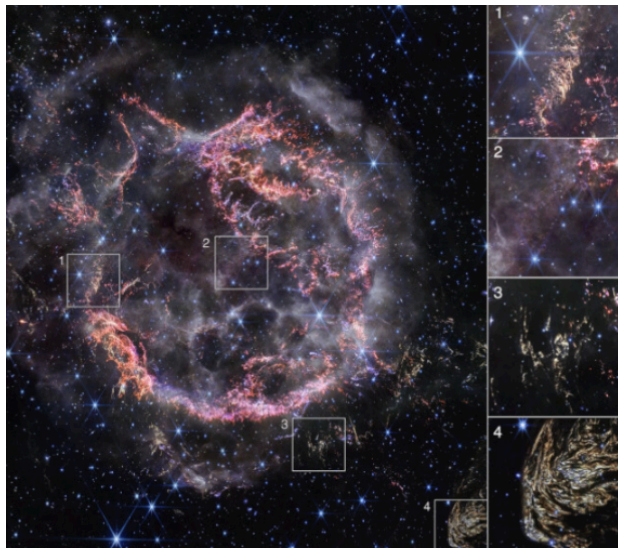
26. (3.00 pts) Identify the DSO in the image:

27. (3.00 pts) This star system has three stars in it. What star is the brightest component of the system and what type of star is it?

28. (3.00 pts) What are 2 characteristics of your answer to 27 (the star type)?

29. (3.00 pts) What is this DSO's associated nebula and what type of nebula is it?

30. (3.00 pts) Which telescope discovered this DSO?



31. (3.00 pts) Identify the DSO in the image and which telescope took the image:

32. (3.00 pts) What feature does Box 1 show?

33. (3.00 pts) Identify the white and purple gas in Box 2 and their cause.

34. (3.00 pts) Box 3 shows dust in the nebula “glowing”. Why does this happen?

35. (3.00 pts) What is the feature in Box 4 often called, and what is it?



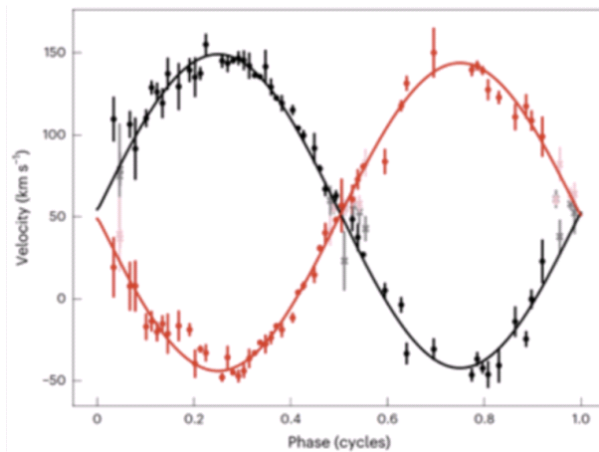
36. (3.00 pts) Identify the DSO in this image:

37. (3.00 pts) What telescope and instrument took this image?

38. (3.00 pts) What is the center object in the image?

39. (3.00 pts) What kind of nebula is this DSO? How can you tell?

40. (3.00 pts) What will this nebula become in ~10 million years?



41. (3.00 pts) Identify the DSO in this image:

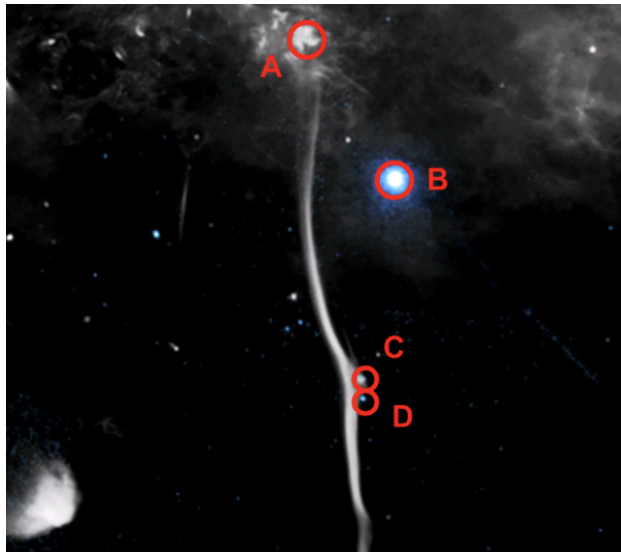
42. (3.00 pts) What method was used above? What's unique about this DSO?

43. (3.00 pts) What does this mean about the mass for each component of the DSO? (Hint: must be under a certain mass limit)

44. (3.00 pts) How close will the objects come within each other (smallest possible orbital time)? How long would the supernova then last for?

45. (3.00 pts)

Let's assume the supernova will occur in 10 billion years and the more massive component is 0.8 solar masses and is gaining mass from the other smaller component. What is the rate at which the more massive component is gaining mass in kg/s?



46. (3.00 pts) Identify this DSO:

47. (3.00 pts) What is this type of structure called? What purpose does horizontal and vertical ones serve?

48. (3.00 pts) What are the blue dots? What telescope captured them?

49. (3.00 pts) What object is causing the break in the center of the filament?

50. (3.00 pts) Where is the object from the above question located in the image?

- ☐ A) A
- ☐ B) B
- ☐ C) C
- ☐ D) D

MATH

51. (3.00 pts)

Let's look at a hypothetical star and planet system. Let's start with the star. Assume that we know the star has $R = 0.1R_{\odot}$ and $\lambda_{\text{peak}} = 700 \text{ nm}$. What is the temperature of the star in Kelvin?

52. (3.00 pts) What is the luminosity of the star in W?

53. (3.00 pts) Using this, obtain luminosity in terms of solar units and get mass in solar units.

54. (3.00 pts) What is the density of this star in kg/m^3 ?

55. (3.00 pts) What is the absolute magnitude of this star?

56. (3.00 pts) If you were observing this system 420.67 pc away, what would the apparent magnitude of the star be?

57. (3.00 pts)

Ignore the above distance now. Let's say we saw an observed wavelength of 666.28 nm of the Hydrogen alpha line (which has a standard wavelength of 656.28 nm). What is the redshift of the star? Is it moving towards or away from us?

58. (3.00 pts) What is the velocity of the star with respect to you in km/s?

59. (3.00 pts) What is the distance to the star now if we use Hubble's Law in Mpc?

60. (3.00 pts) Let's look at the planet now. When we look at a transit, we see that the flux reduces by 1% during transit. What's the radius of the planet in m?

61. (3.00 pts) When we see a transit, the planet doesn't cause an even change in the star's flux as it travels across the observed surface. What phenomenon can explain this?

62. (3.00 pts) The planet is measured to have a transit of ~30 days. What is the distance of the planet to the star in m?

63. (3.00 pts)

Assuming that the universe is expanding at a rate of ~4570 km/s at the place where the system is in respect to you. What is the star's velocity in the frame of the system in m/s?

64. (3.00 pts) What is the planet's velocity in m/s if its mass is much smaller compared to the star?

65. (3.00 pts) What's the minimum mass of the planet in kg?

66. (3.00 pts) What is the incidence flux on the planet from the star in W/m^2 ?

67. (5.00 pts) What is the equilibrium temperature of the planet assuming an albedo of 0?

68. (3.00 pts) Can the planet sustain liquid water?

69. (3.00 pts) Let's look at all of Kepler's Laws and go deeper into them. What are Kepler's Three Laws?

70. (3.00 pts) We can use the First Law to do some calculations with the areas of all ellipses. What's the general form for the total area of an ellipse?

71. (3.00 pts) What would the distances for the periapsis and apoapsis be in terms of the semi-major axis and eccentricity?

72. (4.00 pts) What would the semi-minor axis (b) be in terms of a and e ? What is eccentricity in terms of a and b ?

73. (3.00 pts)

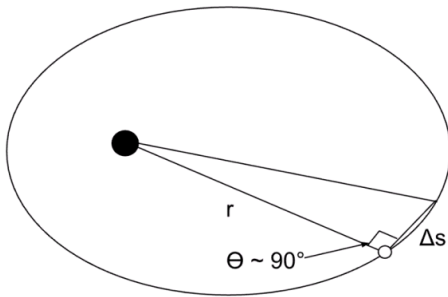
Now, looking at the Second Law, what can we say about the angular momentum of an orbiting body, assuming it's a point mass? (write the equation as a cross product first)

74. (3.00 pts)

What does this equation say about angular momentum, knowing about the nature of ellipses and circles?

75. (5.00 pts)

So, how would the area of an ellipse approximately change with time based on angular momentum and mass, if the law stated earlier applies? Use the following diagram and assume that Δs is a really small distance travelled by the object with some velocity over Δt and can be approximated as a line. Hint: Find what ΔA is equal to as an approximated triangle and then divide by Δt and substitute the equation with L and m .



76. (3.00 pts)

Now, looking at Kepler's Third Law, what force acts on an orbiting body? What net force can this force also be equated to as it orbits in an ellipse/circle? Write their equations too.

77. (4.00 pts)

Knowing that there is only one force and one net force, the two can be equated with each other. When solving for T^2 (period of the orbit squared), what is the equation you get? (this should look familiar)

78. (3.00 pts) What does circular velocity equal to in a circular orbit? What about angular velocity?

79. (3.00 pts)

Write the semi-latus rectum, knowing that it is equal to b^2/a and is the distance to the star when perpendicular to the semi-major axis of an orbit, in terms of a and e .

80. (3.00 pts) What is the equation of ellipse using polar coordinates based on the semi-latus rectum? (distance follows $r = l/(1+e \cos(\theta))$)

81. (3.00 pts) Find the angular momentum of an orbiting object in an elliptical orbit by subbing in v and r at semi-latus rectum.

82. (10.00 pts)

Knowing the value obtained for the change in area over change in time, we can plug in the area of the ellipse over the whole period of the orbit. We can then solve for T^2 and simplify all of the variables and get rid of b and e until we get Kepler's Third Law (woah). Show the derivation of this using this value.